

Solutions to the remaining exercises from the problem sheet

14. Let p be a prime.

(a) Show that $\varphi(p^a) = p^a - p^{a-1}$ for each $a \geq 1$.

Solution. $\varphi(p^a)$ is the number of integers among $0, \dots, p^a - 1$ or, equivalently, among $1, \dots, p^a$, that are not divisible by p . The integers in the latter range (consisting of p^a numbers) that are divisible by p are precisely the p^{a-1} numbers $p, 2p, 3p, \dots, p^{a-1}p$.

(b) Show that $x^p \equiv x \pmod{p}$ for each $x \in \mathbb{Z}$.

Solution. Let $x \in \mathbb{Z}$. If x is not divisible by p , then $x^{p-1} \equiv 1 \pmod{p}$ by Fermat's little theorem and hence $x^p \equiv x \pmod{p}$. On the other hand, if x is divisible by p , then $x^{p-1} \equiv 0 \equiv x \pmod{p}$.

15. (a) Compute $\varphi(5424)$.

Solution. Trial division easily shows that $5424 = 2^4 \cdot 3 \cdot 113$; note that 2, 3, and 113 are primes. Therefore,

$$\varphi(5424) = \varphi(2^4)\varphi(3)\varphi(113) = (2^4 - 2^3)(3 - 1)(113 - 1) = 1792.$$

(b) Show that if $n \geq 2$ is an even integer, then $\varphi(n) \leq \frac{n}{2}$.

Solution. Write $n = 2m$ for $m \geq 1$. Since n is even, any integer x with $0 \leq x \leq n - 1 = 2m - 1$ which is coprime to n has to be odd. There are $m = \frac{n}{2}$ odd integers in this range, namely $1 = 2 \cdot 1 - 1, 3 = 2 \cdot 2 - 1, \dots, 2m - 1$.

16. Let $B \geq 1$ and $k \geq 0$ be integers. For $j = 0, \dots, k$, let a_j be an integer with $0 \leq a_j < B$. Let

$$A = a_0 + a_1B + \dots + a_kB^k.$$

Show that $A \leq B^{k+1} - 1$ with equality if and only if $a_0 = \dots = a_k = B - 1$.

Solution. The claim is clearly true when $B = 1$ so suppose that $B > 1$. The maximal value of A is attained iff $a_0, \dots, a_k$ all take the maximal value $B - 1$. Recall ("geometric series") that $1 + x + x^2 + \dots + x^k = \frac{x^{k+1} - 1}{x - 1}$ for $x \neq 1$. Hence,

$$\begin{aligned} (B - 1) + (B - 1)B + (B - 1)B^2 + \dots + (B - 1)B^k &= (B - 1)(1 + B + \dots + B^k) \\ &= (B - 1) \frac{B^{k+1} - 1}{B - 1} \\ &= B^{k+1} - 1. \end{aligned}$$

17. Recall that a **reduced residue system modulo n** is a set $R \subset \mathbb{Z}$ with the following properties:

- $\gcd(r, n) = 1$ for all $r \in R$.
- For $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$, there exists a unique $r \in R$ with $a \equiv r \pmod{n}$.

Let R be a reduced residue system modulo n and let $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$. Show that $\{ar : r \in R\}$ is also a reduced residue system modulo n .

Solution. Since $\gcd(a, n) = 1$, we have $\gcd(ar, n) = 1$ for each $r \in R$. Let $b \in \mathbb{Z}$ with $\gcd(b, n) = 1$. We need to show that there exists a unique $r \in R$ with $ar \equiv b \pmod{n}$. Let $\bar{a}$ be an inverse of a modulo n . Since $\gcd(\bar{a}b, n) = 1$ and since R is a reduced residue system modulo n , there exists a unique $r \in R$ with $r \equiv \bar{a}b \pmod{n}$. The latter condition is equivalent to $ar \equiv a\bar{a}b \equiv b \pmod{n}$.

18. Find the orders of the different elements of $\mathbb{Z}_7^\times$. Which of these elements are generators of the group?

Solution. First note that $\varphi(7) = 6$ and that $\mathbb{Z}_7^\times$ is cyclic by Gauss's theorem. We first compute the orders of all elements:

- 1 has order 1.
- $2^2 = 4$, $2^3 = 1$ so 2 has order 3.
- $3^2 = 2$, $3^3 = 6$, $3^4 = 4$, $3^5 = 5$, $3^6 = 1$ so 3 has order 6.
- $4^2 = 2$ and $4^3 = 1$ so 4 has order 3.
- $5^2 = 4$, $5^3 = 6$, $5^4 = 2$, $5^5 = 3$, $5^6 = 1$ so 5 has order 6.
- $6^2 = 1$ so 6 has order 2.

Generators are precisely the elements of order 6, i.e. 3 and 5.

19. Alice and Bob use the cyclic group $\mathbb{Z}_{47}^\times$ and its generator $g = 5$ to perform a Diffie-Hellman key exchange. Alice chooses $a = 4$ and Bob chooses $b = 2$. Determine their shared key.

Solution. Their shared key is $5^8 = g^{ab} = (g^a)^b = (g^b)^a$, computed in the group $\mathbb{Z}_{47}^\times$. Repeated squaring modulo 47 gives $5^8 = ((5^2)^2)^2 = (25^2)^2 = 14^2 = 8$.

20. Determine the set of points of the elliptic curve $y^2 = x^3 - x$ over $\mathbb{Z}_5$.

Solution. We first compute the squares in $\mathbb{Z}_5$. These are $0 = 0^2$, $1 = 1^2 = 4^2$, and $4 = 2^2 = 3^2$.

We have $x^3 - x = x(x-1)(x+1) = 0$ if and only if $x = 0, 1, 4$. This gives the points $(0, 0)$, $(1, 0)$, $(4, 0)$ on our elliptic curve. Trying out the different elements of $\mathbb{Z}_5$, we find that $x^3 - x = 1$ if and only if $x = 2$ and that $x^3 - x = 4$ if and only if $x = 3$. This gives the points $(2, 1)$, $(2, 4)$, $(3, 2)$, $(3, 3)$.

In total, the 8 points are $\mathcal{O}$, $(0, 0)$, $(1, 0)$, $(4, 0)$, $(2, 1)$, $(2, 4)$, $(3, 2)$, $(3, 3)$.